INCH-POUND

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5 October 1990
SUPERSEDING
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3 August 1983

MILITARY SPECIFICATION SHEET

BATTERY, RECHARGEABLE, SEALED NICKEL CADMIUM, BB-590/U

This specification is approved for use by LABCOM, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-B-49436

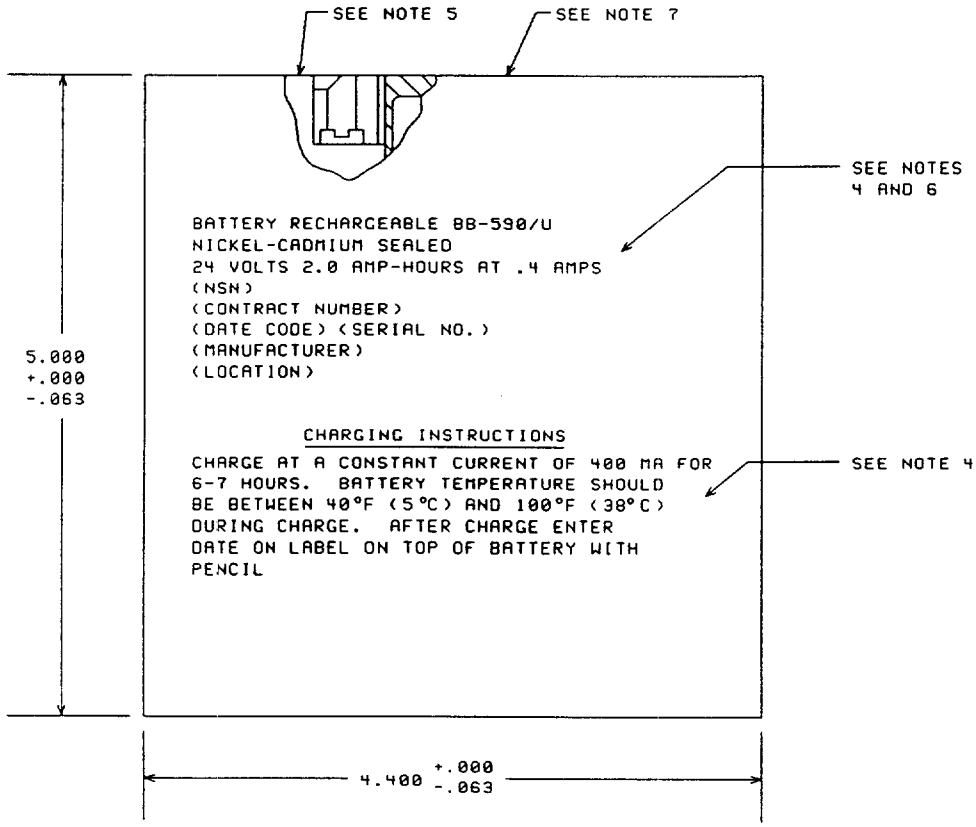


FIGURE 1. FRONT VIEW.

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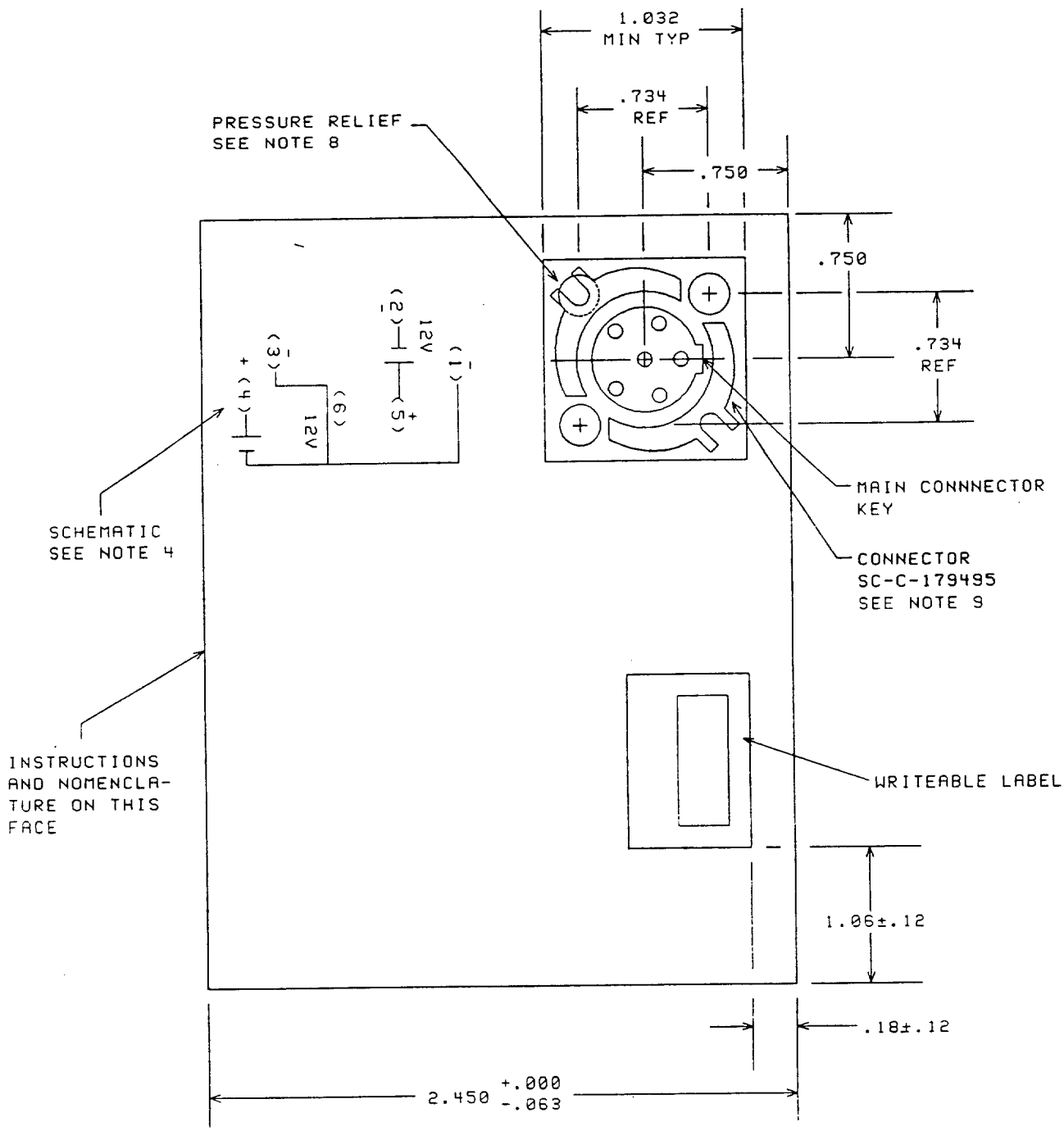


FIGURE 2. TOP FACE OF BATTERY.

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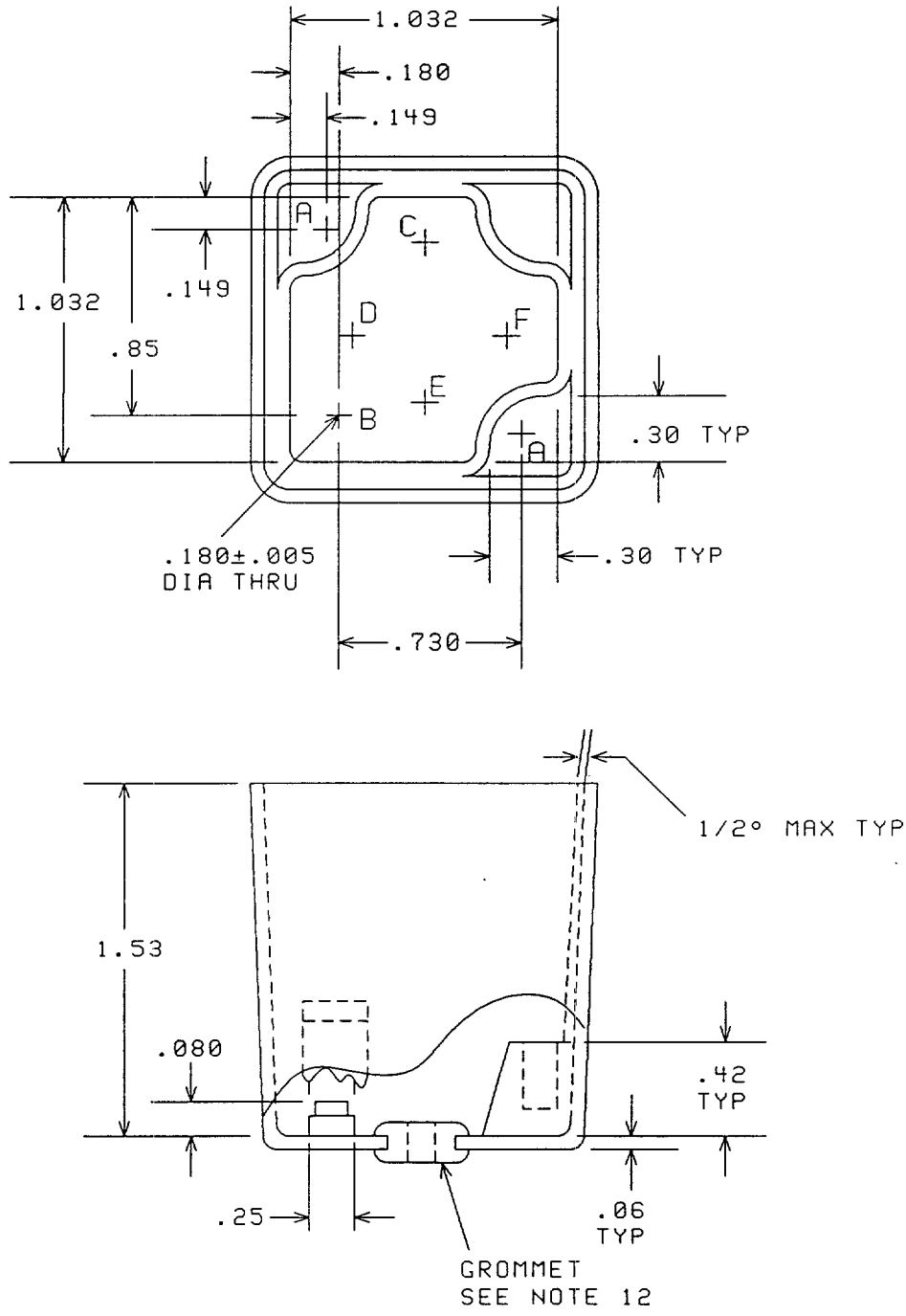


FIGURE 3. BATTERY CONNECTOR INTERFACE

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DRAWING NOTES

1. All dimensions in inches.
2. Tolerances on decimals: ± .031 inches.
3. Dimensioning shall be per USAS-144.5 Y14.5 1966.
4. Stencil (or equivalent) schematic, identification and charging instructions as shown. Color marking white.
5. Face of connector shall be flush to .031 below top of battery.
6. Manufacturer must fill in applicable information in parenthesis.
7. See Figure 2 for top view.
8. Pressure relief valve shall be located under connector. It shall release gas before case damage is incurred.
9. See Figure 3 for connector-battery interface. The main connector key shall be oriented as shown in Figure 2. The two connector mounting screws shall be corrosion resistant steel, Class 303, Condition A in accordance with ASTM-A484.
10. Holes A shall be .154 ±.002 dia x .34 ±.02 deep for a 4-40 UNC-2A insert. Insert to be flush or below surface.
11. The lettered holes of table I are shown in approximate positions in Figure 3. The holes shall be .136 ±.002 diameter. Marking of holes not required.
12. Each required SC-C-179495 connector wire shall be provided with a hole (see table I and figure 3) and pre-fitted with a neoprene rubber grommet of 50 ±5 durometer in accordance with MIL-R-3065 to furnish a watertight seal around the connector wires.
13. Pressure relief valve mounted in hole B has a typical height of .42 from base on connector well.

TABLE I. CONNECTOR WIRE COLOR CODE

HOLE C	HOLE D	HOLE E	HOLE F
BLACK/WHITE	BLACK	WHITE/RED	RED

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REQUIREMENTS:

Rated Capacity (C/5):	2.0 AH
Voltage (Nominal):	24.0 volts
Weight (Maximum):	1.70 Kg (3.75lbs)
Connector:	Floating type per SC-C-179495
Number and type of cells:	20 each sealed "C_SD" size cells or equal
Charge rate for testing and on battery (C/5 for 6-7 hours):	.4 Amps

Discharge Rates	Second Voltage(V)	Final Voltage(V)	Current (Amps)
C/5	-	20.0	0.40
C_5	-	20.0	2.0
$5C_5$	22.0	20.0	10.0
$9C_5$	-	20.0	18.0

Test	Discharge Rate	Capacity Requirement(A)
Full Capacity Discharge	C_5	1.8
Cycle Life, 224 Cycles	C_5	1.44
Overcharge, 24 hours	C_5	1.8
High Rate Discharge	$5C_5$	1.1
Low Temperature Discharge at -30°C (-22°F)	C_5	0.6
Retention of Charge 7 days at 51°C (124°F)	C_5	0.9
Pulse Discharge 18 amps, 5 seconds on, 25 seconds off, continuously to 20 volts	$9C_5$	1.1
Vibration	C/5	2.0

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NOTES: 1. All values are minimum, unless otherwise specified.
2. All charges and discharges shall be performed on fully assembled batteries through the terminals.

Custodian:
Army - ER

Preparing Activity:
Army - ER

Review Activities:
Army - CR, AV, GL

(Project 6140-A719)

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